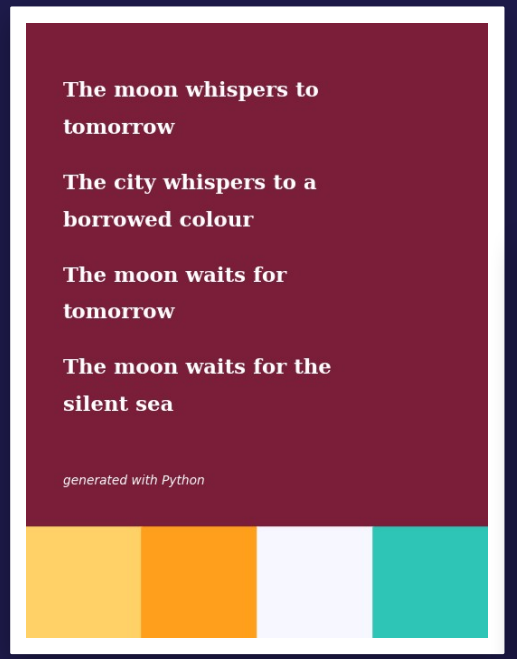
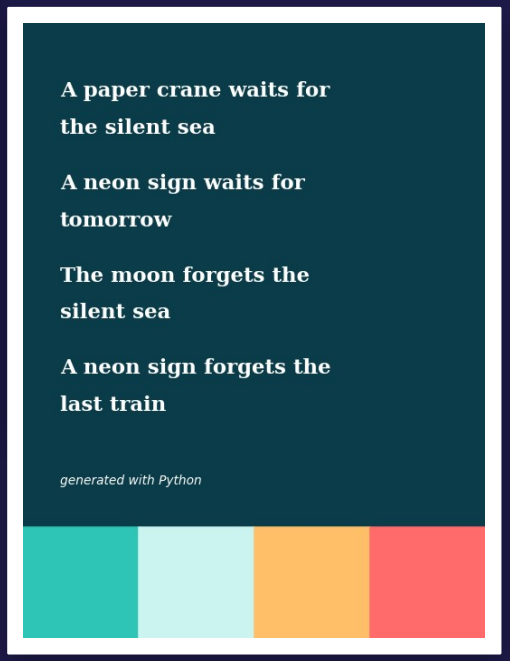
AI FOR CREATIVITY · LEVEL 6 · WEEK 1

What is AI?

Coding as a creative medium

Today you write your first program — and it makes poetry and colour.





Dr. Geetha Mohan · Future Education University College

WEEK 1 AT A GLANCE

Six scheduled hours, one creative thread



2 h

Session A

Technical demonstration

Concepts + live coding: poem and palette generator



2 h

Session B

Lab workshop

Pairs build a generator for their own discipline



1 h

Session C

Group seminar

AI in the creative industries: hype vs reality



1 h

Session D

Tutorial

Skills audit and forming coding pairs

Plus 15 h independent study: 2 h videos · 4 h practice notebook · 3 h reading · 6 h portfolio (R&D journal)

A

SESSION A · TECHNICAL DEMONSTRATION

From 'what is AI?' to your first generative artwork

Concepts first, then we live-code a poem and colour-palette generator together in Google Colab.

2 hours · type along in pairs

HOOK · 15 MIN

Art, tool or threat? You decide.

Refik Anadol

Unsupervised (MoMA, 2022–23): a model trained on the museum's collection dreams new images in real time.

Holly Herndon

Holly+: a musician releases an AI model of her own voice that others can sing through.

Film & VFX

AI now assists rotoscoping, de-ageing and clean-up — hours of manual frame work cut to minutes.

Fashion

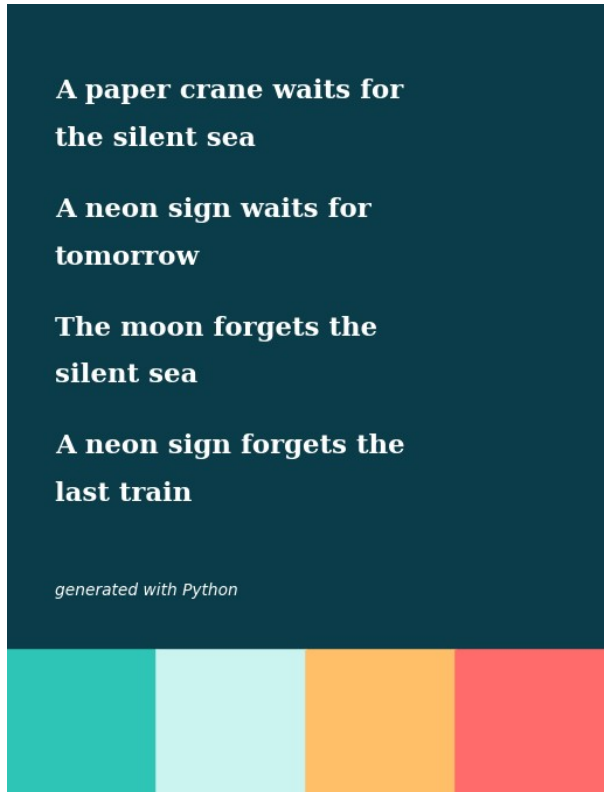
Brands use models to spot emerging trends in millions of social images before they peak.

Quick vote for each: 🎨 art · 🛠️ tool · ⚠️ threat — hands up, or poll on the class channel.

BY THE END OF THIS WEEK

You will be able to…

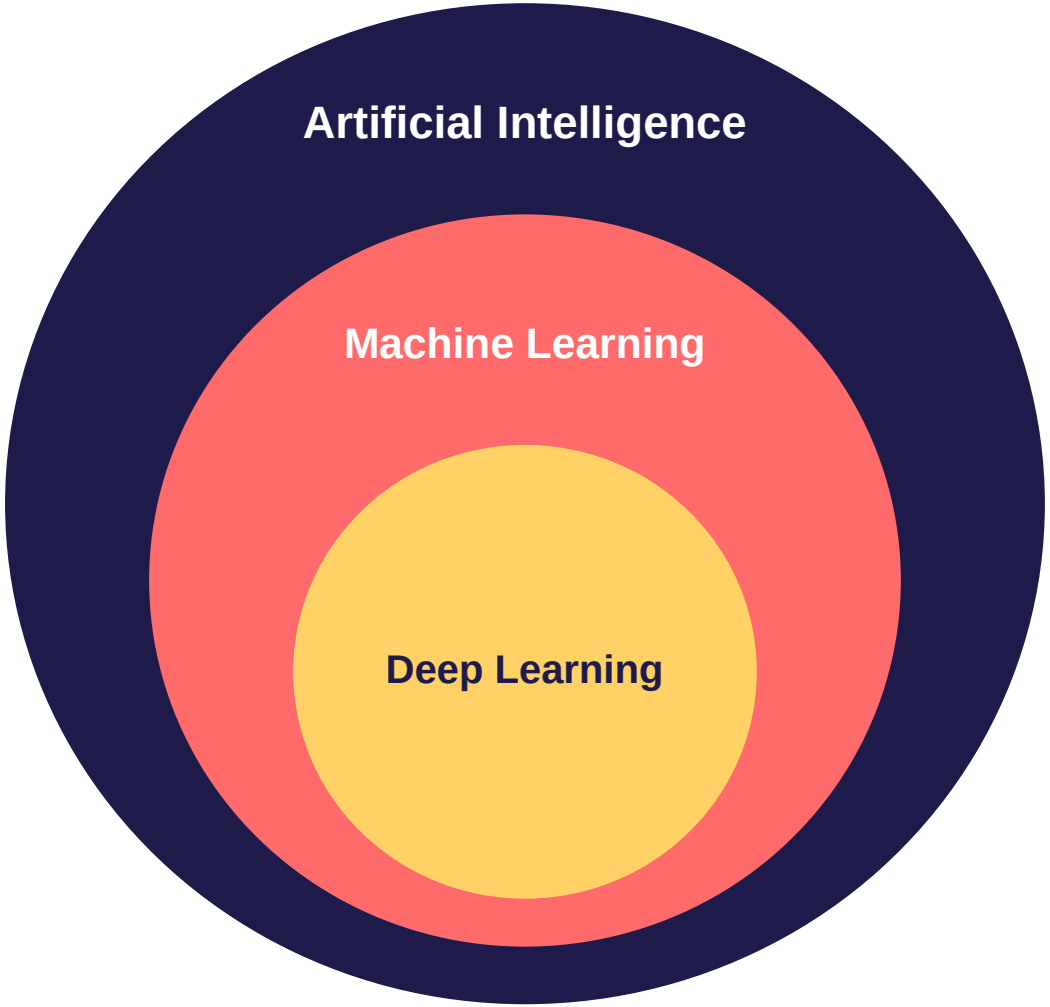
1. Explain AI, machine learning and deep learning in plain words — with a creative example of each
2. Open, run and save a notebook in Google Colab
3. Use variables, strings, lists, random choice and loops to generate text and colour
4. Start your R&D journal: the first page of your portfolio



Your output today

CONCEPTS

AI, ML and deep learning: circles in circles



Artificial Intelligence

Any computer system doing tasks we'd call 'intelligent' if a human did them.

e.g. A chess engine, a chatbot

Machine Learning

AI that learns patterns from examples instead of following hand-written rules.

e.g. Spotify guessing your next song

Deep Learning

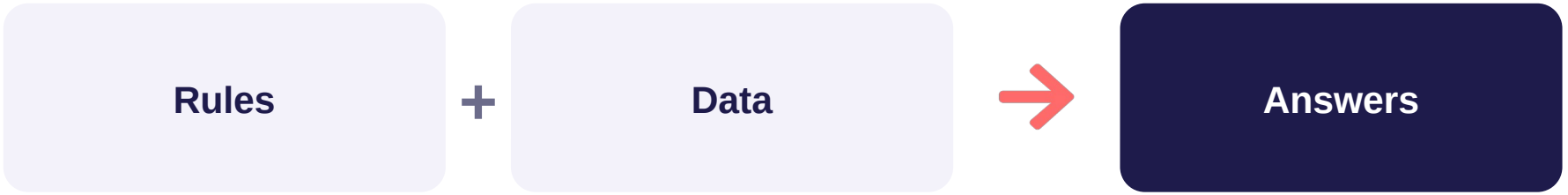
Machine learning with many-layered neural networks — great with images, sound, text.

e.g. Image generators, voice cloning

CONCEPTS

The big shift: from rules to learning

Traditional programming



Machine learning



Creative example

Rules: 'if a painting has lots of blue and swirls, call it Van Gogh.'

Learning: show 1,000 labelled paintings; the model works out the rules itself.

CONCEPTS

Think of machine learning like cooking



Data

the ingredients

Photos, songs, sketches, text, sales records



Model

the recipe it learns

Patterns found by studying the examples



Prediction

the dish it serves

A label, a recommendation, a new image or line of text

Bad ingredients → bad dish. Biased data → biased predictions. (We return to this in Week 3.)

CONTEXT

AI is already in every creative studio

Music

Stem separation, mastering, AI voices

Film & animation

VFX clean-up, storyboards, dubbing

Fashion

Trend forecasting, virtual try-on

Games

Procedural worlds, smarter NPCs

Graphic design

Background removal, layout ideas

Architecture

Generative massing, render assist

YOUR STUDIO

Google Colab: a coding studio in your browser

1. Go to colab.research.google.com and sign in with Google
2. File → Upload notebook → Week01_Demo_Poem_and_Palette.ipynb
3. Click a cell, press Shift + Enter to run it
4. File → Save a copy in Drive — your work is kept

Why Colab first?

- Nothing to install — works on any laptop
- Free GPU for heavier AI work later
- Same notebook file (.ipynb) opens in Jupyter
- Week 4: we install Anaconda + JupyterLab for local work

LIVE CODE 1 · VARIABLES

Variables are labelled boxes

In [ ]:

```
artist = "Vera"
medium = "code"
year = 2026

print(f"{artist} works in {medium}")
print(f"Exhibiting in {year}")
```

Out:

```
Vera works in code
Exhibiting in 2026
```

In plain English

- = means 'put this in the box'
- Text in quotes is a string
- Numbers need no quotes
- f"...{name}..." drops a variable into text
- **Try: change artist and medium to describe you**

LIVE CODE 2 · LISTS AND CHANCE

Lists + chance = your first generative system

In [ ]:

```
import random

subjects = ["the moon", "a paper crane", "the city"]
verbs    = ["whispers to", "forgets", "paints"]
objects  = ["the silent sea", "tomorrow", "the last train"]

print(subjects[0])   # Python counts from 0
print(random.choice(subjects), random.choice(verbs),
      random.choice(objects))
```

Out:

```
the moon
the city paints the last train
```



Chance as a creative tool

Composer John Cage used chance operations to make decisions in his music. Dada poets pulled words from a hat.

random.choice() is our hat.

LIVE CODE 3 · LOOPS

Loops repeat with variation

In [ ]:

```
for line_number in range(4):
    line = f"{random.choice(subjects)} {random.choice(verbs)} {random.choice(objects)}"
    print(line.capitalize())
```

Out:

```
The city forgets the silent sea
A paper crane whispers to tomorrow
The moon paints the last train
The city whispers to the silent sea
```

Watch the indentation

The 4 spaces tell Python which lines belong inside the loop. range(4) means 'do it 4 times'.

LIVE CODE 4 · PACKAGE IT

Wrap it up: a poem generator you can reuse

```
In [ ]:
def make_poem(n_lines=4):
    """Return a list of randomly generated poem lines."""
    poem = []
    for _ in range(n_lines):
        s = random.choice(subjects)
        v = random.choice(verbs)
        o = random.choice(objects)
        poem.append(f"{s} {v} {o}".capitalize())
    return poem

poem = make_poem()
```

New words

- def — define a reusable recipe
- append — add to a list
- return — hand back the result

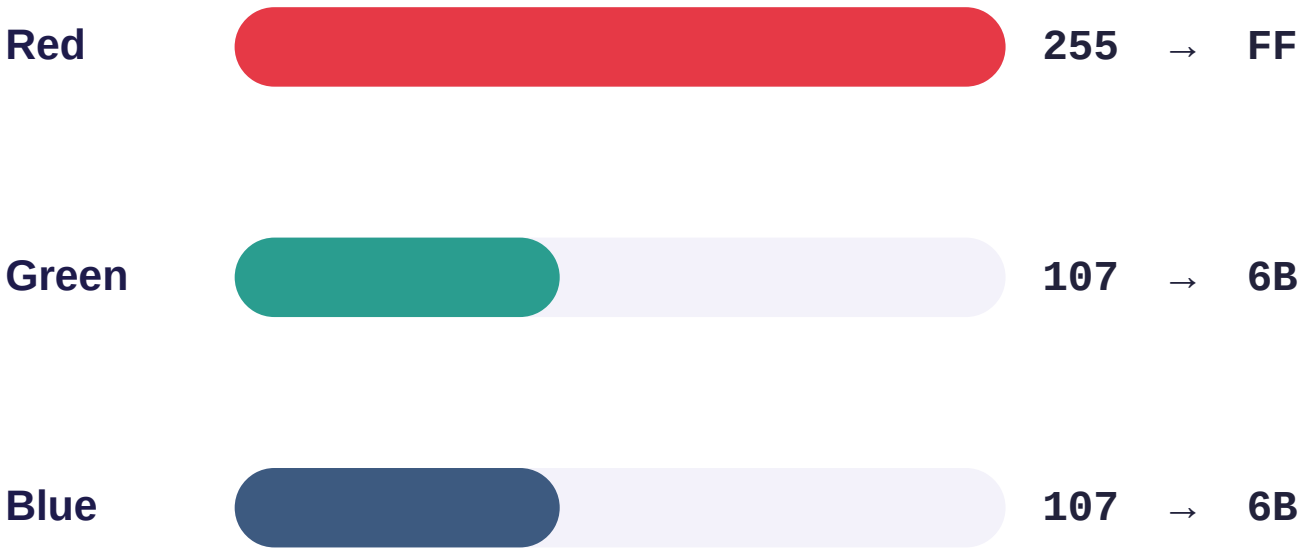
Full detail next week

One word — make_poem() — now gives us a new poem every time.

LIVE CODE 5 · COLOUR

Colour is numbers too

#FF6B6B



Every screen colour = Red + Green + Blue light, each 0–255.

```
In [ ]:
r, g, b = 255, 107, 107
hex_code = f"#{r:02X}{g:02X}{b:02X}"
print(hex_code)   # #FF6B6B
```

LIVE CODE 6 · DESIGN THE CHANCE

Artists don't just use chance — they constrain it

Any colour

```
def random_colour():
    r = random.randint(0, 255)
    g = random.randint(0, 255)
    b = random.randint(0, 255)
    return f"#{r:02X}{g:02X}{b:02X}"
```

Warm only

```
def warm_colour():
    r = random.randint(200, 255)   # lots of red
    g = random.randint(60, 180)    # some green
    b = random.randint(0, 90)      # little blue
    return f"#{r:02X}{g:02X}{b:02X}"
```



#3A7FC1 #B2E83D #5C1F8A #E8D4C0 #12A67B



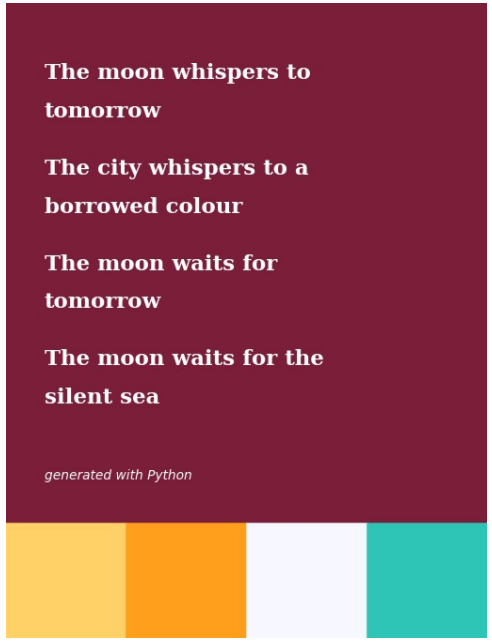
#F2743A #E0A13B #FF5A3C #D9822B #F39C4F

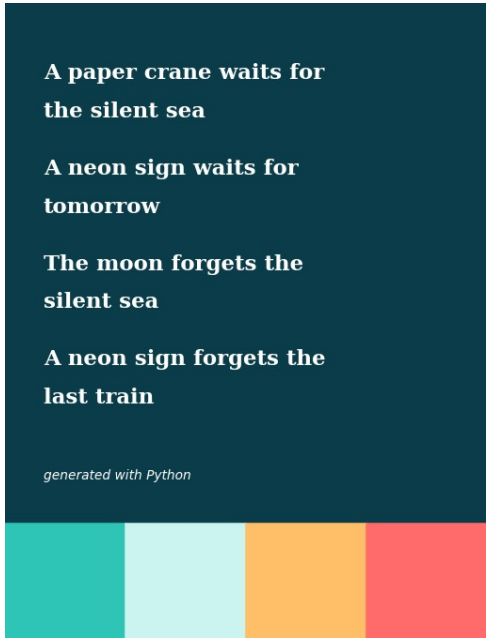
You write the rules. The machine explores the possibilities.

LIVE CODE 7 · BRING IT TOGETHER

Poem + palette = a generative poster







These three were made by the code in today's notebook.

```
In [ ]:
make_poster(make_poem(),
  ["#1E1B4B", "#FF6B6B",
   "#FFD166", "#2EC4B6",
   "#F7F7FF"])
```

Same code, three runs

Each poster is unique. It saves as my_poster.png — download it for your R&D journal.

ERRORS ARE NORMAL

Red text is feedback, not failure

SyntaxError

```
print("missing a quote)
```

Python can't read the line — check quotes and brackets

NameError

```
print(colour_palette)
```

You used a name that doesn't exist — check spelling

IndentationError

```
for i in range(4):
print("hi")
```

Lines inside a loop need 4 spaces

Golden rule: read the LAST line of the error first.

EXIT TICKET · 10 MIN

“One thing AI could do for my discipline is…”

One sentence on the class channel before the break. We'll read a few aloud in the seminar.

B

SESSION B · LAB WORKSHOP

Build a generator for your own discipline

Work in pairs. Start from today's demo code and make it yours — then show it in a gallery walk.

2 hours · Week01_Lab_Build_Your_Generator.ipynb

YOUR BRIEF

Make a generator only your discipline would make

Fashion

Slogan or collection-name generator

Film

Film titles and one-line loglines

Perfume & branding

Scent names with a matching palette

Games

Magic items, spells, planet names

Music

Band names, album titles, lyric seeds

Photography

Shot-list prompts for a street walk

Output for the gallery walk: 10 generated ideas + a themed palette + one poster.

TWO TRACKS

Choose your track: Core or Stretch

🟢 CORE

- Replace every FILL_ME in the notebook
- Write 3 word lists (5+ words each) for your theme
- Generate 10 ideas with a loop
- Set min/max RGB values for a themed palette
- Save week1_poster.png

🔵 STRETCH

- Build a dictionary of moods → words + colour ranges
- Write generate(mood) returning a poem and palette
- Use a while loop and input() so users keep asking
- Handle unknown moods with if … not in …
- Bonus: a haiku mode that counts syllables

LAB RUNNING ORDER

Two hours, five moves

0:00 — 1

Pair up

Mixed confidence pairs; open the lab notebook

0:10 — 2

Build

Core or Stretch; tutor circulating

1:10 — 3

Break

10 minutes

1:20 — 4

Gallery walk

Every pair shows its best output

1:50 — 5

Save & share

Drive + class channel link

Gallery walk prompt: leave one sticky note per screen — 'I like…', 'I wonder…', 'What if…'

SESSION C · GROUP SEMINAR · 1 H

AI in the creative industries: hype vs reality

Reading
du Sautoy, M. The Creativity Code, Chapter 1

Warm-up (5 min): read aloud three exit-ticket answers from Session A

1. **Is a program that surprises its maker creative?**
2. **Which tasks in your discipline would you happily hand to AI — and which never?**
3. **Who owns an artwork made by a generator you wrote?**

SESSION D · TUTORIAL · 1 H

Skills audit: where are you starting from?

Rate yourself 1 (new to this) → 5 (confident)



Coding 1 2 3 4 5



Maths 1 2 3 4 5



Data 1 2 3 4 5



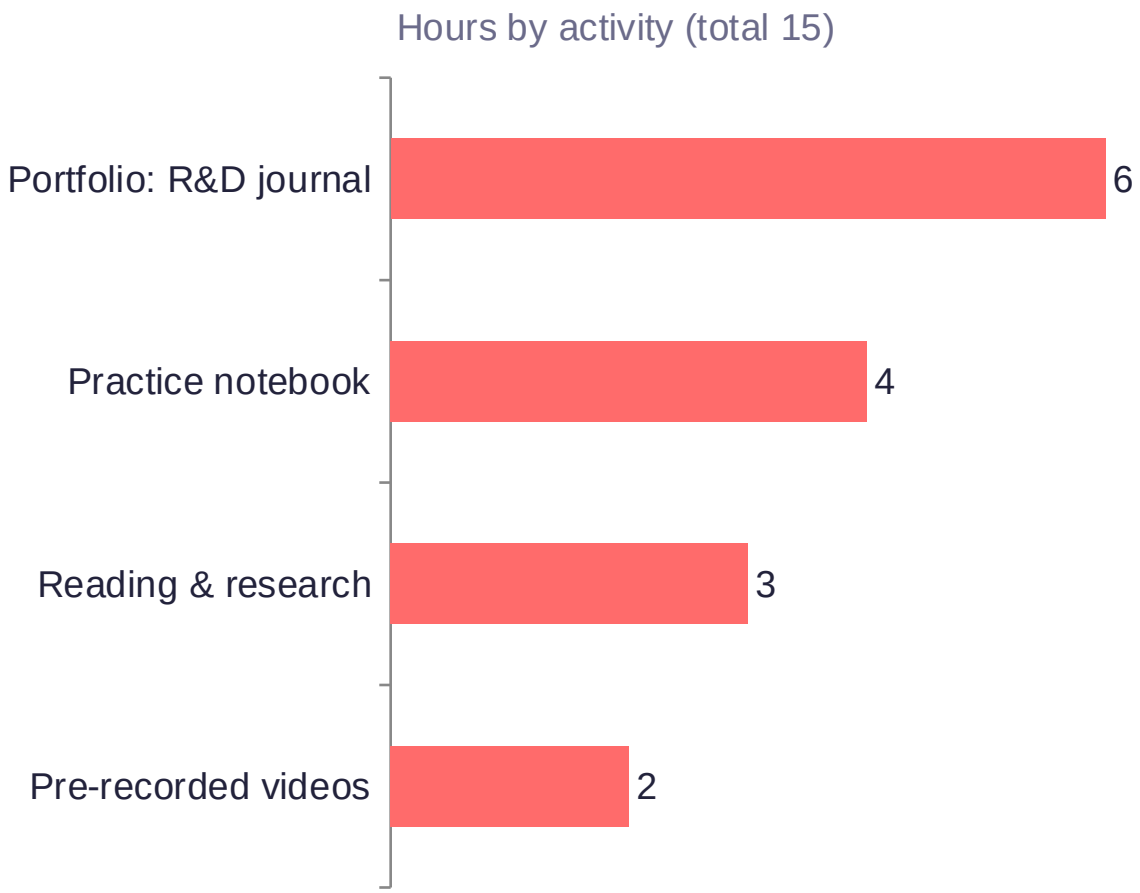
My creative discipline 1 2 3 4 5

What we do with it

- Form balanced coding pairs
- Flag who needs extra support early
- Re-rate in Week 7 and Week 14 to see growth

BEFORE WEEK 2 · 15 H INDEPENDENT STUDY

Your week between sessions



To do

- Watch Week 2 videos: functions, if/else, images are numbers
- Finish the 10 practice tasks in the Week 1 notebook
- Read The Creativity Code, Ch. 1; find 3 AI-art case studies
- R&D journal: case studies + poster + 300-word reflection
- **Bring 3 of your own photos or artworks next week**

You wrote the rules.

The machine explored the possibilities.

Next week: every image is secretly a grid of numbers — and we'll make glitch art from them.